Strict Parser Replay: Correcting Measurement, Not Training

Experiment index

Question and Context

Could permissive parsing execute a valid-looking part of an otherwise invalid model response, overstating benchmark performance?

Change and Method

Reparse saved raw generations with strict-complete-response-v1, replay fake tool execution, and write separate result files. No model inference or training was rerun.

Malformed calls, invalid meta-tool fields, conflicting decisions and trailing output reject the whole response. There is no accepted valid subset or automatic repair.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
100/100/100 BAYMAX / strict replay	150	42/150 (0.280)	0.500	0.694	0.013	0.172
Schema-diverse / strict replay	150	40/150 (0.267)	0.480	0.672	0.025	0.172
Historical ACT200 / strict replay	150	46/150 (0.307)	0.493	0.738	0.000	0.086
Historical ACT150 / strict replay	150	46/150 (0.307)	0.480	0.764	0.025	0.052

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
100/100/100 BAYMAX / strict replay (150 cases)	0.367	0.113

Evaluation	Macro-F1	Invalid-output rate
Schema-diverse / strict replay (150 cases)	0.395	0.133
Historical ACT200 / strict replay (150 cases)	0.349	0.120
Historical ACT150 / strict replay (150 cases)	0.308	0.107

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Result Provenance

- 100/100/100 BAYMAX / strict replay: 2026-09-28 UTC, 150 cases.
- Schema-diverse / strict replay: 2026-09-28 UTC, 150 cases.
- Historical ACT200 / strict replay: 2026-09-28 UTC, 150 cases.
- Historical ACT150 / strict replay: 2026-09-28 UTC, 150 cases.

Interpretation

Full-suite success counts changed from 43 to 42 for ACR, 42 to 40 for schema-diverse, 47 to 46 for ACT-balanced and stayed 46 for ACT-midpoint.

These differences correct the measuring instrument. They must not be counted as model regressions caused by new training.

Provenance and Limits

Replay artifacts retain source/scenario hashes and inherit the original generation cost/latency. That latency cannot measure replay performance or a new model-speed gain.

Old split leakage and unseeded LoRA initialization were not repaired by replay. Source filenames embedded in JSON can predate renaming; use FILE_RENAMES.json to locate their current paths.

The linked parser README documents rejected-output counts and the corrected synthetic split. Later routing metrics may be derived from saved decisions without changing these historical JSON files.

Supporting Files

- README.md
- FILE_RENAMES.json
- SCORING.md

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.